

# Cells, and how they look in klayout, xschem and magic

This is a walk through what the PDK actually contains - the generated cell library - and what each of the four tools shows you when you open it. It also covers how to activate the PDK and how to move between it and the open-source PDKs already installed in the container.

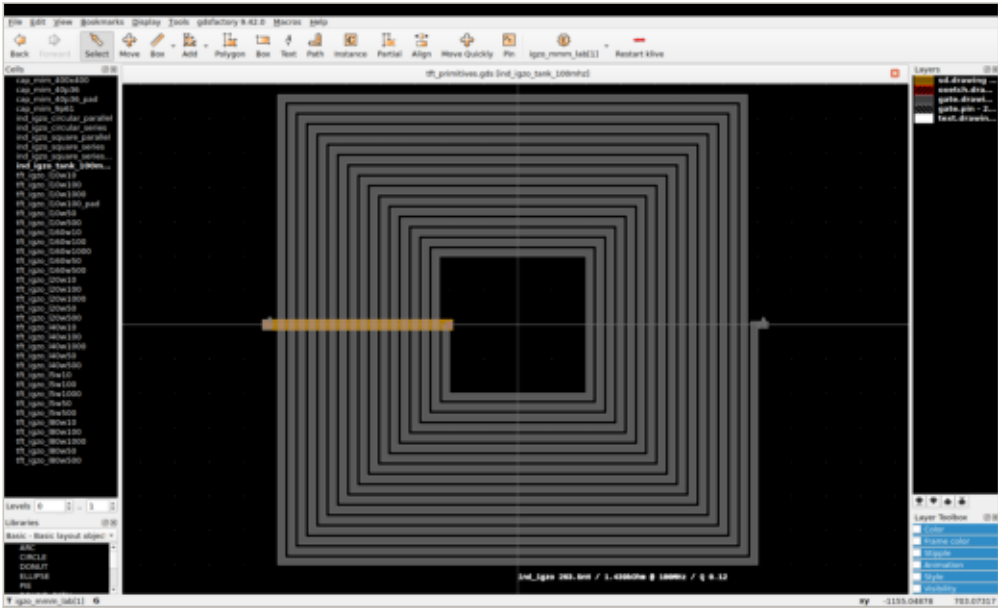
## The process in one paragraph

A top-gate coplanar IGZO thin-film transistor, four masks. The IGZO island goes down first, the Au source/drain lands directly on it - the contact is the overlap, there is no via layer - a blanket 50 nm Al2O3 follows, opened only over the probe pads, and the Au gate is the last metal.

| # | Layer  | GDS | What it is                                            |
|---|--------|-----|-------------------------------------------------------|
| 1 | igzo   | 4/0 | semiconductor island, deposited first                 |
| 2 | sd     | 6/0 | Au source/drain, directly on the IGZO                 |
| 3 | oxetch | 5/0 | openings in the blanket Al2O3, over the S/D pads only |
| 4 | gate   | 2/0 | Au gate, last metal, above the dielectric             |
|   | align  | 8/0 | alignment marks                                       |

## Where everything lives

|                 |                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------|
| Repository      | <a href="https://github.com/Juander28/TFT-MMM-LAB-PDK">github.com/Juander28/TFT-MMM-LAB-PDK</a> |
| Working copy    | /foss/designs/TFT-MMM-LAB-PDK (= C:\TFT-MMM-LAB-PDK)                                            |
| Installed as    | \$PDK_ROOT/TFT-MMM-LAB-PDK, with PDK_ROOT = ~/pdks                                              |
| This document   | ~/Documents/PDK-TFT-MMM-LAB                                                                     |
| Technology name | igzo_mmm_lab                                                                                    |
| Device library  | igzo_mmm_lab_pr                                                                                 |



The 39 generated cells in KLayout, technology `igzo_mmm_lab` selected. Cell list on the left, layer palette on the right.

# Activating the PDK

Once, after cloning:

```

$ cd /foss/designs/TFT-MMM-LAB-PDK
$ ./install.sh
$ source ~/.bashrc           # or just open a new terminal

```

install.sh does two things. It builds a writable PDK\_ROOT at ~/pdks holding a symlink to every PDK in /foss/pdks plus this one, and it adds a use-pdk function to ~/.bashrc. Nothing is copied and the vendor PDKs are untouched.

## Why not simply put it in /foss/pdks

/foss/pdks belongs to root and the designer account cannot write there, so a new directory cannot be created next to sky130A and gf180mcuD. With a root shell in the container one symlink makes it native, and install.sh does it automatically whenever /foss/pdks happens to be writable:

```

ln -s /foss/designs/TFT-MMM-LAB-PDK /foss/pdks/TFT-MMM-LAB-PDK

```

## Switching PDKs

Then, in any terminal:

```

$ use-pdk TFT-MMM-LAB-PDK      # the IGZO TFT process
$ use-pdk gf180mcuD           # GlobalFoundries 180 nm
$ use-pdk sky130A              # SkyWater 130 nm
$ use-pdk ihp-sg13g2           # IHP SiGe BiCMOS 130 nm
$ use-pdk ihp-sg13cmos5l       # IHP CMOS variant

```

One command switches all four tools at once: they all read the same four variables. use-pdk only makes that a single command instead of four exports.

| Variable           | Set to                                 | Who reads it                 |
|--------------------|----------------------------------------|------------------------------|
| PDK                | TFT-MMM-LAB-PDK                        | everything, by name          |
| PDKPATH            | <i>PDK_ROOT</i> /PDK                   | scripts, .magicrc, xschemrc  |
| KLAYOUT_PATH       | ~/.klayout:\$PDKPATH/libs.tech/klayout | klayout: technology + PCells |
| SPICE_USERINIT_DIR | \$PDKPATH/libs.tech/ngspice            | ngspice: model search path   |

## What each tool picks up

|         |                                                                        |
|---------|------------------------------------------------------------------------|
| klayout | technology igzo_mmm_lab, its layer colours, the igzo_mmm_lab_pr PCells |
| xschem  | \$.:IGZO_MODELS, the tft_igzo symbol, the example testbenches          |
| ngspice | design.ngspice (device wrappers) and the corner sections               |
| magic   | igzo_mmm_lab.tech, the .mag device views, extraction rules             |
| netgen  | libs.tech/netgen/setup.tcl for LVS                                     |

# The cell library

Three PCells generate everything. The library ships 39 cells: the transistor matrix drawn on the test chip, the same device with its probe pads, three capacitors and five inductors. The passives have their own document - the cell SOP - which goes through every parameter and where its default came from.

| Cell                  | Count | What it is                                                    |
|-----------------------|-------|---------------------------------------------------------------|
| tft_igzo_l{L}w{W}     | 30    | L in {5,10,20,40,80,160} um x W in {10,50,100,500,1000} um    |
| tft_igzo_l10w100_pad  | 1     | the fabricated device, 400x400 um probe pads included         |
| cap_mim_400x400       | 1     | the overlap capacitor Cox was measured on                     |
| cap_mim_40p36 / _9p61 | 2     | sized by capacitance instead of by dimensions                 |
| ind_igzo_*            | 5     | square/circular x series/parallel, plus the 100 MHz tank coil |

## They are generated, not drawn

There is one description of the geometry - the KLayout PCell - and every other view is derived from it, so they cannot drift apart:

```
$ python3 scripts/build_libs_ref.py      # PCells -> gds/tft_primitives.gds
$ ./scripts/build_mag_views.sh          # that GDS -> mag/*.mag (32 magic views)
```

Do not hand-edit the .mag files; they are output, not source.

## The defaults are the fabricated device

tft\_igzo with its default parameters reproduces cell TFT-10L100W of the test chip shape for shape, on all three drawn layers. That is asserted by a regression test, not by eye:

```
$ python3 libs.tech/klayout/tech/pymacros/testing/test_pcell_vs_gds.py \
    libs.ref/igzo_mmm_lab_pr/gds/TFT_layout.gds

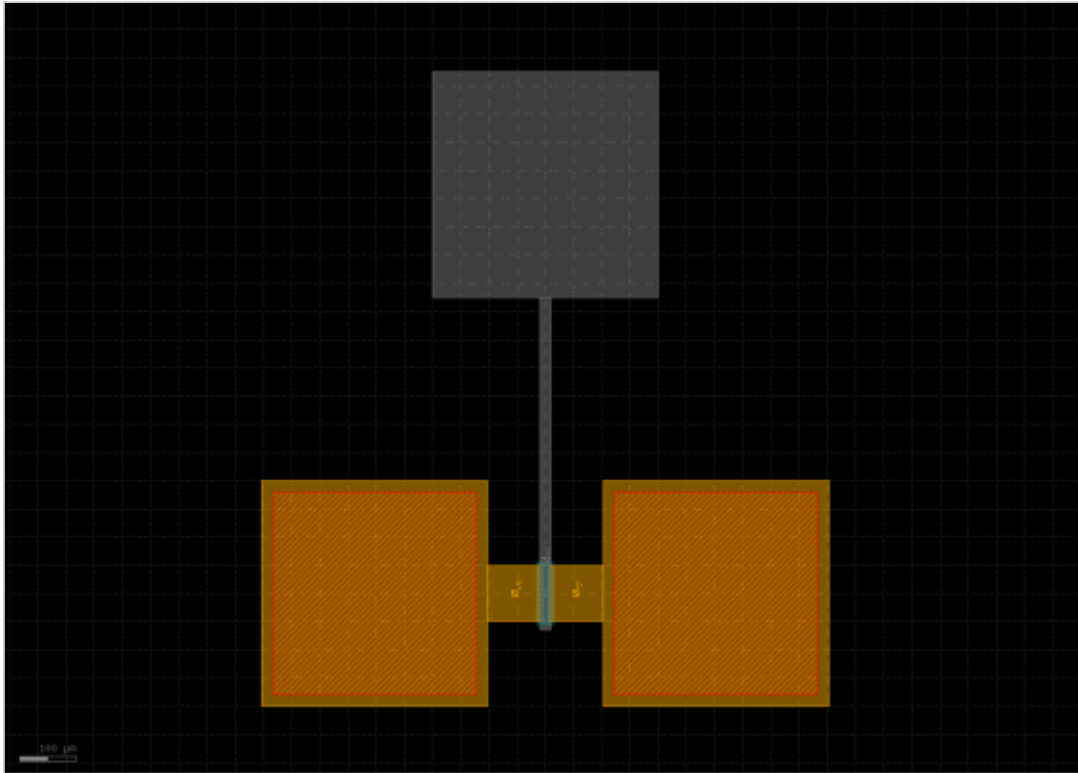
OK sd 6/0    [(-502.5, -200.0, -5.0, 200.0), (5.0, -200.0, 502.5, 200.0)]
OK igzo 4/0  [(-15.0, -55.0, 15.0, 55.0)]
OK gate 2/0  [(-200.0, -495.0, 200.0, 495.0)]
OK gate 2/0  PCell stripe 20 um over the island, overlap [5.0, 5.0] um per side
OK gate 2/0  GDS   stripe 20 um over the island, overlap [5.0, 5.0] um per side
OK nf=1/2/5  shared electrodes, W_total = 100 / 200 / 500 um
OK drawn channel gap 10 um - the +3 um print bias is reported, not drawn
```

## Naming

l10w100 means L = 10 um, W = 100 um, both drawn. The process prints the channel about 3 um longer; the PCell reports that as l\_printed and never applies it to the geometry, because a PCell writes a mask and the bias is one sample of one cell.

# In KLayout

Library `igzo_mmm_lab_pr` appears in the library browser with two PCells. Drop one in and edit its parameters in the instance dialog.

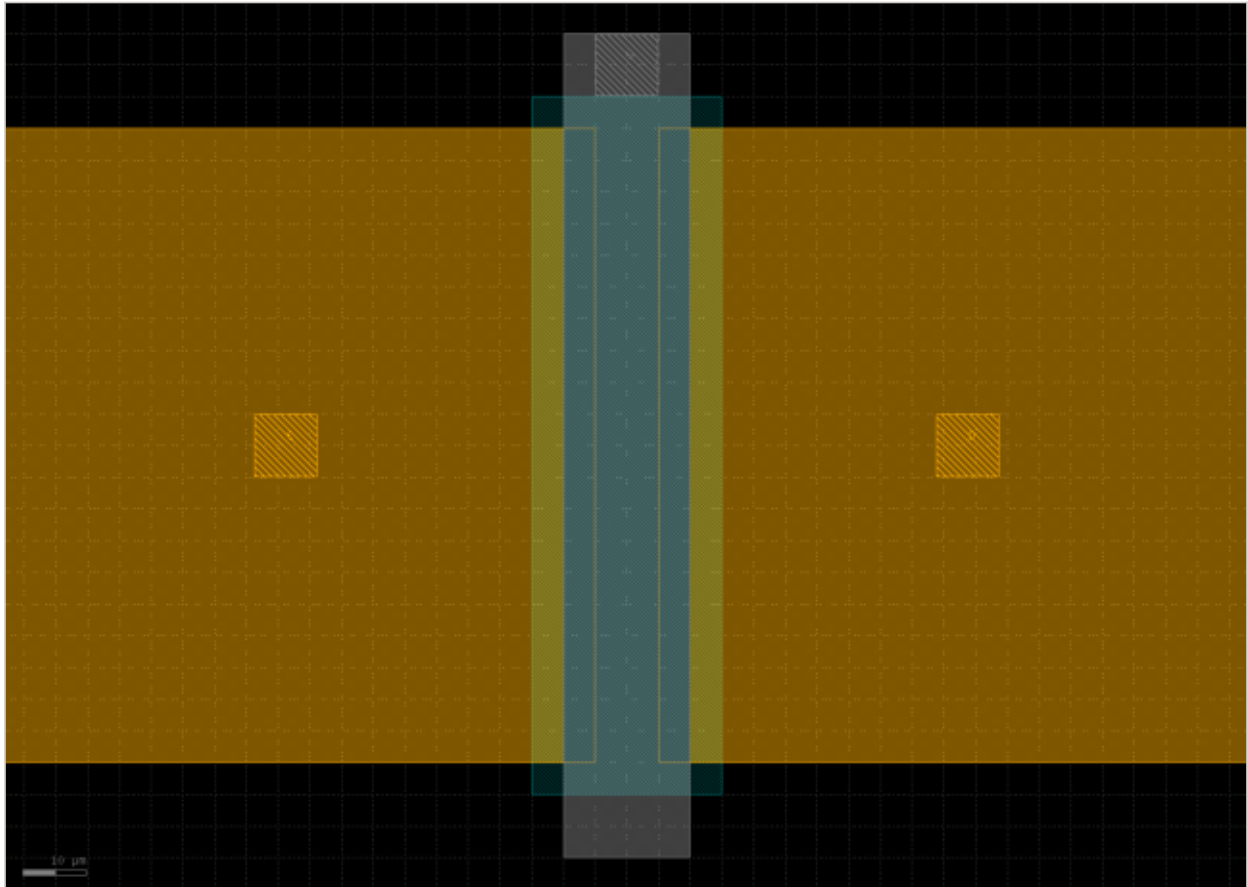


*tft\_igzo with pad=true: S/D pads left and right (orange, with the red dielectric openings inside them), gate pad above, device in the middle. This is the fabricated TFT-10L100W.*

## tft\_igzo parameters

| Parameter                    | Default   | Meaning                                        |
|------------------------------|-----------|------------------------------------------------|
| <code>w_gate</code>          | 100 um    | channel width, per finger                      |
| <code>l_gate</code>          | 10 um     | drawn channel length = the electrode gap       |
| <code>nf</code>              | 1         | fingers; they share electrodes                 |
| <code>gate_ov</code>         | 5 um      | gate overlap per side - the dominant parasitic |
| <code>sd_len</code>          | 97.5 um   | S/D electrode length                           |
| <code>igzo_ext_x / _y</code> | 10 / 5 um | island over the electrode / past its end       |
| <code>pad</code>             | off       | 400x400 um probe pads + dielectric openings    |
| <code>l_bias</code>          | 3 um      | drawn-to-printed bias, reported only           |
| <code>l_printed</code>       | read-only | $l\_gate + l\_bias$                            |
| <code>w_total</code>         | read-only | $w\_gate \times nf$                            |

## In KLayout: the device itself

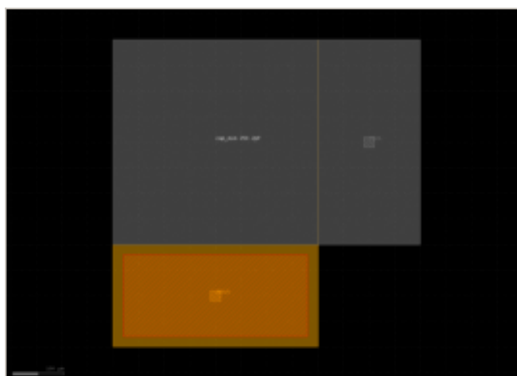


The channel region of tft\_igzo\_l10w100. Teal is the IGZO island, orange the S/D electrodes, grey the gate stripe crossing them. The gap between the two orange electrodes is  $L$ ; the height over which they face the island is  $W$ .

Read off the picture, and identical to what the test chip draws:

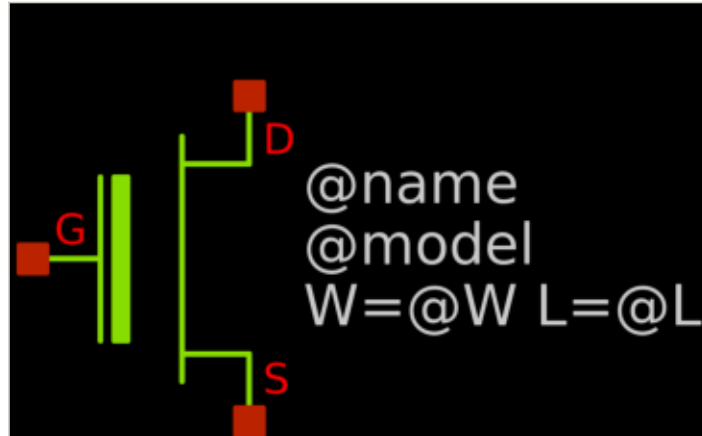
|                                               |                                                                                                 |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------|
| $L = 10 \text{ } \mu\text{m}$                 | the gap between the S/D electrodes                                                              |
| $W = 100 \text{ } \mu\text{m}$                | the electrode height, not the island height                                                     |
| overlap = $5 \text{ } \mu\text{m}$            | how far the gate reaches over each electrode                                                    |
| island = $30 \times 110 \text{ } \mu\text{m}$ | $10 \text{ } \mu\text{m}$ over each electrode in x, $5 \text{ } \mu\text{m}$ past each end in y |

### The capacitor



cap\_mim\_400x400: the lower plate is S/D metal running down, the upper plate is gate metal running sideways, and the capacitance is their  $400 \times 400 \text{ } \mu\text{m}$  crossing -  $250.2 \text{ pF}$  measured, which is where  $C_{ox} = 1.564 \text{ fF}/\mu\text{m}^2$  comes from.

# In xschem



symbols/tft\_igzo.sym. Three pins in the order D, G, S - the same order the igzo\_tft subcircuit declares. There is no bulk pin: a TFT sits on glass, and the wrapper keeps the intrinsic device's bulk tied to its own source.

## What it netlists to

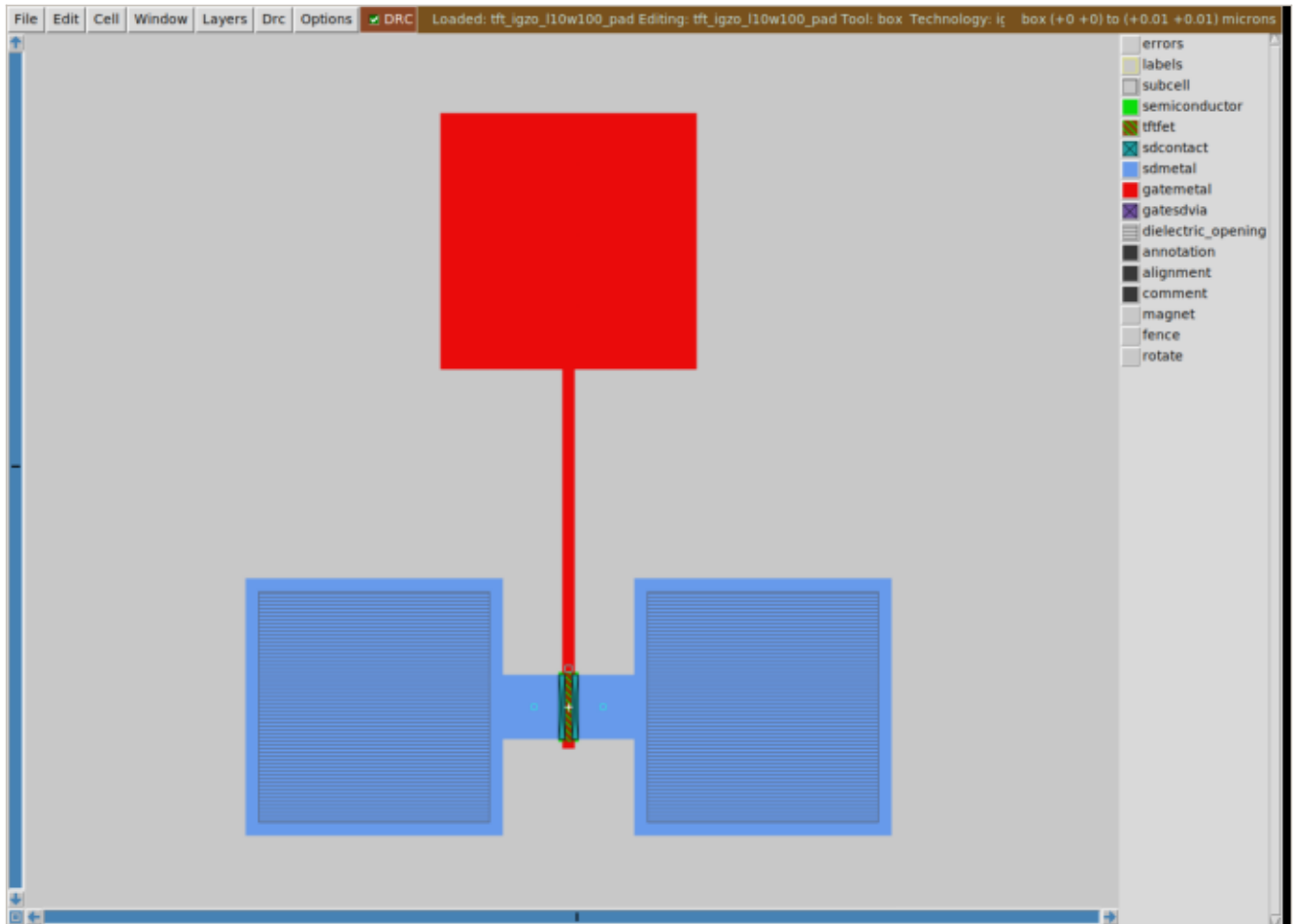
```
XM1 D G S igzo_tft W=1000u L=8u ov=5u nf=1
```

W, L and ov are drawn dimensions in metres. spiceprefix=X because the device is a subcircuit: the wrapper adds the measured contact resistance and the overlap capacitance around a level-1 core, so ov can be swept - which is the point, since the overlap is what decides whether a tuned stage works.



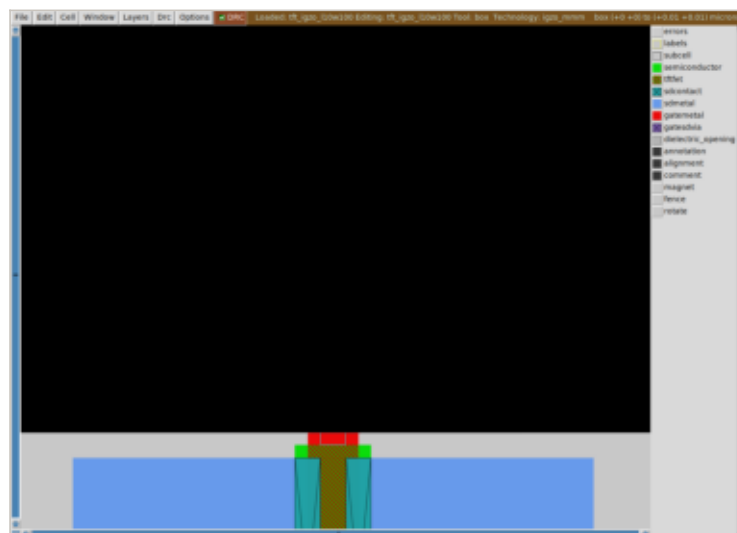
tests/tft\_iv.sch - the output-characteristics testbench on the device the model was validated against ( $W = 1000 \text{ um}$ ,  $L = 8 \text{ um}$ ). The MODELS block pulls in the wrappers and one corner; the NGSPICE block holds the sweep.

# In magic



*tft\_igzo\_l10w100\_pad in magic. Blue is sdmetal, red gatmetal, and the hatched squares inside the S/D pads are the dielectric openings. The layer palette on the right is this process, not a MOS one.*

## How magic decomposes the device



*Zoomed in: blue sdmetal electrodes, teal sdcontact where the Au meets the IGZO, the hatched tftfet channel between them, green semiconductor stubs above and below - the island overhang - and red gatmetal top and bottom.*

# In magic: what comes out

---

## Extraction

```
% load $PDKPATH/libs.ref/igzo_mmm_lab_pr/mag/tft_igzo_l10w100
% extract all
% ext2spice lvs
% ext2spice -o layout.spice

.subckt tft_igzo_l10w100 S D G
X0 D G S igzo_tft w=0.11333m l=10u
.ends
```

$l$  comes out exactly: 10  $\mu\text{m}$ , the gap between the electrodes.  $w$  does not, and the picture on the previous page shows why - the extractor measures the gated island, and the green stubs are drawn 5  $\mu\text{m}$  past each electrode for alignment margin. They are gated but carry no source-to-drain current, so the device width is the electrode height, 100  $\mu\text{m}$ , while extraction reports about 13 % more.

This is a property of how the test chip is drawn, not a bug in the deck. An island flush with the electrodes extracts exactly. Until then netgen is told to allow it.

## LVS

```
$ netgen -batch lvs "layout.spice tft_igzo_l10w100" \
                  "schematic.spice tft_igzo_l10w100" \
                  $PDKPATH/libs.tech/netgen/setup.tcl

Circuit 1 contains 1 devices, Circuit 2 contains 1 devices.
Circuit 1 contains 4 nets,    Circuit 2 contains 4 nets.

Final result:
Circuits match uniquely.
```

setup.tcl permutes source and drain, holds  $l$  to 1 % and allows 20 % on  $w$ , and says in the file itself why the two tolerances differ. Setting  $W$  wrong in the schematic does fail - it was checked:  $W = 500\text{u}$  against the extracted 113  $\mu\text{m}$  reports property errors rather than passing quietly.

## Three terminals, on both sides

Magic emits three terminals because the msubcircuit line in the techfile carries no substrate arguments. The other form, "device subcircuit", also emits three - but in the order G S D, which would disagree with the symbol silently. Pin order is the first thing LVS checks, and a swapped gate is the one mistake that still simulates.



# In ngspice

Two lines pull in the process. The first is the device wrappers, the second a corner:

```
.include PDK_ROOT/PDK/libs.tech/ngspice/design.ngspice
.lib      PDK_ROOT/PDK/libs.tech/ngspice/igzo_mmm_lab.ngspice tt
```

```
XM1 d g s igzo_tft W=100u L=10u ov=5u
```

In xschem the same two lines read `$::IGZO_MODELS/...` - the variable the PDK's xschemrc sets.

## Corners

| Section | What it is                                                                   |
|---------|------------------------------------------------------------------------------|
| tt      | typical: median over every working device of the best chip                   |
| best    | long-channel devices of the best chip, not contact-limited                   |
| short   | $L \leq 10 \text{ um}$ , contact-limited; prefer tt plus the wrapper instead |
| all     | across all three chips, including the bad ones - worst case                  |

## What the wrapper adds

|                     |                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------|
| Contact resistance  | $R_c = 3.3 \text{ Ohm} \cdot \text{m} / W$ per contact, measured from $R_{on}$ against $L$ |
| Overlap capacitance | $C_{ox} \cdot ov \cdot W$ per side; 7.82 fF per um of width at $ov = 5 \text{ um}$         |
| An inert bulk       | b is held through a 1 Tohm resistor; the core ties its own bulk to source                  |

$C_{gso}$  and  $C_{gdo}$  are zero in the model cards on purpose. The overlap capacitance is not a property of the process alone - it is  $C_{ox}$  times the drawn overlap - so it lives in the wrapper where it tracks  $ov$ , instead of being frozen at the 5 um of the test chip.

## The number that says the model still works

```
W = 1000 um, L = 8 um, VGS = 6 V, VDS = 10 V
  simulated    654.4 uA
  measured     670 - 701 uA
```

That is the same figure the original extraction reported, and `run_checks.sh` asserts it, so a change to the models that breaks it fails loudly.

# The four tools agree

```
$ use-pdk TFT-MMM-LAB-PDK
echoPDKPATH
/headless/pdks/TFT-MMM-LAB-PDK
$ ./scripts/run_checks.sh
```

run\_checks.sh runs the whole chain and stops at the first failure:

| Check                             | Tool    | Result                        |
|-----------------------------------|---------|-------------------------------|
| PCell against the fabricated cell | klayout | identical on all three layers |
| Technology and library discovery  | klayout | igzo_mmm_lab, igzo_mmm_lab_pr |
| DRC on the generated primitives   | klayout | clean                         |
| Symbol netlists with D G S        | xschem  | pin order correct             |
| Model against the measurement     | ngspice | 654.4 uA                      |
| GDS round-trip                    | magic   | layer-exact                   |
| Device extraction                 | magic   | one igzo_tft, l = 10 um       |
| Layout against schematic          | netgen  | match uniquely                |

## What the PDK does not know

1. The models are DC-only. They reproduce the measured output curves - median simulated/measured = 1.01 over 18 devices - and nothing else has been checked. Cox is measured, but no C-V or S-parameter measurement backs the AC behaviour.
2. Level 1 imposes a square law; the measured saturation exponent is 2.31, so the model runs about 20 % pessimistic on gm. Valid to VGS ≤ 6 V, VDS ≤ 10 V.
3. The +3 um print bias is one sample. The PCell reports it, never applies it.
4. The DRC deck is preliminary. Every rule is measured or read off the test chip; there is no foundry rule document. 5 um is the minimum drawn dimension because the smallest drawn device is 5 um and it prints at 8 um.
5. TFT2 is not represented: its saturation exponent is 0.87, which no square-law model can express.

## Running the deck over the fabricated chip

|                       |                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------|
| The transistor matrix | clean, except the 5 um cells - they are drawn with 3.5 um of gate overlap instead of 5 um |
| Test_capacitance      | clean                                                                                     |
| Test_mobility         | violates several rules - it is a measurement structure, drawn before any rules existed    |

None of these are reasons not to use the PDK. They are the things a PDK that looks authoritative should not hide.